

Parth Natekar

DEPARTMENT OF ENGINEERING DESIGN, IIT MADRAS

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EDUCATION	INSTITUTION & DEGREE	CGPA	YEAR
	Indian Institute of Technology Madras <i>Bachelor of Technology in Engineering Design,</i> <i>Master of Technology in Biomedical Design</i>	8.89/10.00	2020
	Fergusson College <i>Higher Secondary School</i>	92.4 %	2015
	Vikhe Patil Memorial School <i>Secondary School (CBSE)</i>	97.0 %	2013

PUBLICATIONS	Parth Natekar , Avinash Kori, and Ganapathy Krishnamurthi. "Demystifying Brain Tumour Segmentation Networks: Interpretability and Uncertainty Analysis." <i>Frontiers in Computational Neuroscience</i> 14 (2020): 6 (Link). Avinash Kori, Parth Natekar , Balaji Srinivasan, and Ganapathy Krishnamurthi. "Abstracting Deep Neural Networks into Concept Graphs for Concept Level Interpretability." <i>arXiv preprint arXiv:2008.06457</i> (2020). (Link).
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RESEARCH & PROFESSIONAL EXPERIENCE	Anheuser-Busch InBev, Bangalore Data Scientist Machine Learning for Compliance Analytics <i>Aug '20 - Present</i> - Working on building machine learning models and developing hypotheses to flag anomalous and risky transactions for legal and financial compliance.
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Medical Image and Reconstruction Lab, IIT Madras

Research Associate | Interpretable Deep Learning for Medical Image Analysis

(Guide: [Dr. Ganapathy Krishnamurthi](#), *Engineering Design, IIT Madras*) *June '19 - June '20*

- Conceptualized a pipeline for interpretability and uncertainty estimation of deep learning models which perform various image processing tasks in the medical domain, such as tumor segmentation.
- This involves implementing techniques like Network Dissection and Bayesian Dropout, and extending techniques like Activation Maximization and Grad-CAM to segmentation models.
- Implemented the pipeline on benchmark brain tumor segmentation datasets to understand information flow and representation in internal layers and to quantify uncertainty of model predictions.
- Developed a python package for direct implementation of the proposed interpretability pipeline on deep neural networks for any relevant biomedical dataset.

Honeywell Technology Solutions, Bangalore

Research Intern | Graphical Modelling of Neural Networks for Causal Inference

(Mentor: [Satyanarayan Kar](#), *Honeywell Aerospace, India*) *January '19 - June '19*

- Devised a probabilistic graphical model-based algorithm for Causal Inference on a deep neural network, to provide human-understandable explanations of what the network actually learns.
- Implemented the algorithm on benchmark datasets and successfully delineated variables in the model which were causal for a particular prediction compared to those with spurious correlations.
- Developed a deep learning pipeline for time-series analysis of flight data in order to predict anomalous flight landings. Achieved 86% accuracy in predicting instability 2 minutes prior to landing.
- Implemented an algorithm for object detection and text extraction from aircraft documents, automating the electronic file conversion of thousands of files.

C-Design Lab, Purdue University, USA

Research Intern | Wearables for Human-Computer Interaction

(Guide: *Dr. Karthik Ramani*, Mechanical Engineering, Purdue University) May '18-July '18

- Completed the entire design cycle – ideation, detail design, fabrication and prototyping, of IOT based flexible, wearable haptic devices for interaction with Mixed Reality environments.
- Devised a novel method to design flexible electronic circuits on various fabric substrates. Conducted material and sensor tests to integrate tactile sensing using carbon based piezoresistive sensors.
- Developed multiple ergonomic, working prototypes. Set up networking capabilities via UDP/TCP protocols and tested the wearables in Mixed Reality and IoT environments.
- Created applications in Unity and C# to validate the efficiency of using the prototyped wearables.
- Reduced task time by 44% and error rate by 74% vis-à-vis traditional interaction methods.

CERN (European Organization for Nuclear Research), Geneva

Engineering Intern | Testing Apparatus for CMS cooling system

(Mentor: *Antti Onnela*, *EP-DT*, CERN, Geneva) May '17 - July '17

- Selected to be part of a 3-member team from IIT Madras to work on the Large Hadron Collider.
- Designed an adiabatic testing apparatus for the evaporative CO² based cooling system of the CMS Detector using concepts from thermal, mechanical and electrical design.
- Set up sensing and feedback system for monitoring of performance of the cooling system
- Created a software interface in LabView to enable easy user interaction.

SCHOLASTIC ACHIEVEMENTS

- **PURE fellow:** Awarded PURE Scholarship for a two-month Research Internship at Purdue University, Indiana, having been selected in the top 14 among 800 students from IIT Madras.
- **CBSE Class X School Topper:** Came first in school with percentage of 97.00%. Awards for highest marks in Math (100/100) and Science (99/100).
- One of 15 students selected from 5000 by Infosys for their ‘Spark – Catch Them Young’ Program.

RELATED COURSE WORK

Biomedical Design

- Design of Diagnostic and Monitoring Systems
- Design of Implantable and Surgical Devices
- Medical Image Analysis
- Medical Equipment Dissection Lab
- Computational Systems Biology
- Human Anatomy, Physiology & Biomechanics
- Biomaterials Engineering

Mechanical Engineering

- Design of Mechanical Systems (I & II)
- Engineering Mechanics
- Design of Thermal and Fluid Systems
- Manufacturing Processes
- Advanced Materials and Processes
- Geometric Modelling and CAD

Computation and Machine Learning

- Machine Learning for Engineering Applications
- Probabilistic Graphical Modelling
- Computational Systems Biology
- Electromagnetic Compatibility for Design
- Computational Methods in Design
- Introduction to Computation and Visualization
- Programming and Data Structures in Python

Electrical Engineering

- Design of Electronic Systems I
- Digital Signal Processing
- Design of Electronic Systems II (Mechatronics)
- Electromagnetic Compatibility for Design
- Bio-Micro-Electrical-Mechanical Systems
- Signals and Systems

SKILLS

Programming: Python, C, C++, C#, JavaScript, Unity, Machine Learning (Keras, Tensorflow, Pytorch), Arduino, Raspberry Pi, L^AT_EX
Languages: Competent French, Limited Working German.

RELEVANT
COURSE
PROJECTS

Computational Modelling of Pathogen Interactomes

January to May 2018 | Supervisor : [Dr. Karthik Raman](#)

- Delineated drug-resistance pathways in the pathogen Staph Aureus through systems level computational modelling of the protein-protein interaction network of the organism.
- Devised a modified network-based shortest path algorithm by integrating node centrality measures with chemical protein interaction scores between drug molecules and proteins.
- Achieved 68.5% prediction accuracy on already known resistance pathways, validating the viability of our algorithm for exploring drug resistance pathways in pathogens.

Colour Sensing Robotic Manipulator

January to May 2017 | Supervisor : [Dr. N. J. Vasa](#)

- Designed and implemented a 6-dof color detecting robotic manipulator with a magnetic tooltip arm.
- Involved electronic circuit design with the use of a microcontroller to control the robot.
- Coded in C/C++ using the Arduino IDE.

Ultrasound Nerve Image Segmentation

Jan to May 2017 | Supervisor : [Dr. Ganapathy Krishnamurthi](#)

- A deep learning-based project for semantic segmentation of ultrasound nerve cell images.
- Used a fully-convolutional neural-network implemented in Python using Keras and Tensorflow.
- Obtained 71.2% Dice Score on test dataset – on par with then state-of-the-art techniques

TEACHING
AND
VOLUNTARY
EXPERIENCE

Teaching Assistant - Design of Electronic Systems Lab, IIT Madras

Professor : [Dr. Tuhin Subhra Santra](#)

Aug '19 - Dec '19

- Selected as Graduate Teaching Assistant for the Design of Electronic Systems Lab (Fall 2019).
- Managed and guided a group of 60+ students in electronic circuit design principles.

Teaching Assistant - Medical Image Analysis, IIT Madras

Professor : [Dr. Ganapathy Krishnamurthi](#)

Aug '19 - Dec '19

- Conducted tutorial sessions to teach students the programming aspects of Medical Image Analysis.

Deputy Head, International Relations Cell, IIT Madras

July '17 - July '18

- Led and organized a team of 20+ people, managed a group of 80+ international exchange students.
- Helped organize various events, including 'International Day', which catered to 2000+ people.

Volunteering

- Worked with an organization called *Lata Mahajan Foundation* on their project *EducationLinked*, which is aimed at overall development of tribal and underprivileged children.
- Worked with *iVolunteer*, Bangalore to help organize health camps in orphanages.

EXTRA
CURRICULAR
ACTIVITIES

Institute Football Team

Aug '16 - Dec '17

- One of only two freshers to be a part of the Institute Football Team.
- Captain of School Football Team. Participated in CBSE West Zone Tournament in 2011 and 2012

Literary and Debate

- Secured the 3rd position for Ganga Hostel, IIT Madras in the Lit-Soc inter-hostel debate competition.
- Part of organizing committee of IIT Madras Model United Nations 2017.

Art and Design

- Selected Impressionist paintings uploaded on [Behance](#).
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